Tuning Into Emotion: A Comparative Study of Model Accuracy, Precision, and Recall on Raw and Preprocessed Singing Data

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Abstract—**Advancements in affective computing have increasingly explored emotion recognition from speech; however, singing represents a uniquely complex modality due to its melodic, rhythmic, and lyrical intricacies. This study investigates how audio preprocessing influences the performance of bimodal models that fuse audio and textual data for emotion recognition in Filipino singing. Twenty natural singing recordings were collected in a real-world acoustic environment and divided into raw and preprocessed datasets, the latter undergoing noise reduction and spectral balancing via FL Studio 25's Edison plugin and the FabFilter suite. Unlike large-scale deep learning approaches, this study utilized robust machine learning classifiers suitable for limited data: Random Forest and Support Vector Machine (SVM). Acoustic features were extracted using Librosa, while lyric features were processed using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization with linguistic descriptors. Model predictions were generated using a feature concatenation strategy and validated through Leave-One-Out Cross-Validation. The impact of preprocessing on model performance was statistically evaluated through paired t-tests and McNemar's tests. By focusing specifically on Filipino music, this study addresses a significant gap in culturally diverse emotion recognition research and aims to provide practical insights for developing accurate Multimodal Emotion Recognition Systems for music with limited training data.**

Keywords—Speech Emotion Recognition, Audio Preprocessing, Bimodal Models, Late Fusion, Term Frequency–Inverse Document Frequency

I. Introduction

Computing has evolved from creating machines that perform tasks to developing systems that can understand, interpret, and respond to human emotions, a field known as affective or empathetic computing (Billinghurst, 2025). A fundamental modality explored in empathetic computing is that of human voice, a common yet rich channel for human expression (Pinheiro, 2025). As a result, Emotion Recognition in Speech (ERS) has become a foundational area of research wherein deep learning models are able to demonstrate success in recognizing emotion from spoken language (Lian et al., 2023).

However, human emotional expression is not limited to speech. Music and singing present a more nuanced form of emotional expression and presents a specialized scope that dives recognizing emotion from music and singing. In contrast to conventional speech, recognizing emotion in singing presents a series of challenges because singing is governed by musical structures such as melody, rhythm, harmony, and lyrics which are employed in such a way that helps vocally express emotion (Liyanarachichi et al., 2025).

The complex nature of singing, therefore, calls for a multimodal approach where the lyrical meanings and vocal performance are explored through textual and audio modalities respectively. Fusing the data extracted from these sources remains a challenging task for developing accurate emotional recognition systems for singing (Liyanarachichi et al., 2025). In addition, the cultural influences and styles behind different music genres can carry unique characteristics that require further research and focus.

To develop a system that accurately recognizes emotion through singing would be fundamentally dependent on a model trained on quality data. Real-world singing data is often imperfect and is subject to variable recording conditions, environmental contexts, and other audio artifacts (Awan et al., 2024). To address the issue, data preprocessing techniques like noise reduction, filtering, and normalization serve as an essential and necessary step in its development, especially for machine learning (Sharma et al., 2020). Despite this, cleaning raw audio data can still introduce risks of altering or eliminating completely subtle audio signals such as vocal wavers or breathy tones, which can also serve as examples of emotional expression.

The primary focus of this study is to experimentally determine the impact of audio preprocessing on the performance of a bimodal (audio-text) emotion recognition model specifically tailored for Filipino singing. While preprocessing techniques are often assumed to be beneficial, their effect on the complicated, melodic, and dynamic nature of sung vocals is not well-established (Hao et al., 2025). To explore this, the study is guided by the following research question: *"How does applying noise reduction and audio normalization to singing data affect the accuracy, precision, and recall of a bimodal emotion recognition model compared to using raw, unprocessed data?"*

The main hypothesis of this study is that the model trained on the preprocessed dataset will demonstrate a statistically significant improvement in overall accuracy, precision, and recall compared to the model trained on the raw dataset. This research aims to provide several key contributions in the field of empathetic computing. First, it will offer observed evidence on the direct impact of data preprocessing in the specialized domain of Singing Emotion Recognition (SER), a topic that is less explored than traditional speech analysis.

The findings will provide a quantitative answer to whether the resource investment in meticulous data cleaning delivers a worthwhile improvement in model performance. Second, the study will offer a methodological contribution by providing practical guidance for researchers and developers on efficient data handling strategies for SER systems. Finally, by focusing on Filipino music, this work begins to address the lack of culturally diverse datasets highlighted in recent literature (Liyanarachchi et al., 2025), giving a unique analysis that improves the inclusivity of music emotion recognition research.

II. Review of Related Literature

A survey paper by Kamma et al. (2023) provides a thorough overview of machine learning techniques applied to music emotion recognition and therapy while emphasizing the efficacy of models such as Random Forest, Convolutional Neural Networks (CNN), and hybrid Support Vector Machine-Artifical Neural Network (SVM-ANN) models. It notes that Random Forest often achieves higher accuracy than CNNs, with feature extraction methods like Mel-Frequency Cepstral Coefficients (MFCC) being commonly employed. The integration of multimodal data, including audio and lyrics, as well as facial expression analysis, enhances emotion detection accuracy in musical applications. Additionally, the paper also explored how present research suggested that music therapy can benefit from emotion recognition systems to tailor therapeutic interventions effectively. The authors cited challenges such as the need for extensive training data and preprocessing for noise reduction (Kamma et al., 2023), which aligns with the significance of investigating preprocessing effects on model performance.

Further solidifying this foundation is a comprehensive review by Patil et al. (2025) which addresses the complex field of music emotion recognition, this review presents a detailed analysis of technologies and methodologies used to analyze emotional content in music. Central to the discussion is the arousal-valence model, which classifies emotions based on physiological activation and emotional positivity or negativity. The authors emphasize the role of machine learning and deep learning techniques such as Support Vector Machines (SVMs), Random Forests, CNNs, and Recurrent Neural Networks (RNNs), in extracting meaningful patterns from the pitch, rhythm, timbre, lyrics, and metadata of audio to be processed. They identify critical gaps, including dataset limitations, difficulties in generalizing models across diverse populations and music genres, and the need for more granular and specific emotion recognition. Future research directions involve improving model interpretability, and developing scalable multimodal systems for applications for adaptive music therapy and affective computing.

Addressing the need for research across diverse music genres and populations, as highlighted by Patil et al., an earlier study investigated the specific domain of Filipino music. In this 2017 study, the researchers utilized Paul Ekman's six fundamental emotions as the framework for categorizing emotional content in songs across distinct Filipino genres such as kundiman, novelty, pop, and rock. The study employed jAudio for extracting musical features, while lyrics features were obtained using a Bag-of-Words representation. For classification, algorithms including Naive Bayes, SVMs, and k-Nearest Neighbors were tested on a dataset of 100 Filipino songs. Findings indicated that models using combined audio and lyric features achieved classification accuracy ranging from 52% to 69%, with Naive Bayes and k-Nearest Neighbors performing better given the limited dataset size (Noblejas et al., 2017). The study contributes valuable insights into emotion recognition within the culturally specific context of Filipino music.

The use of combined audio and lyric features in the study of Filipino music highlights a border trend in steps toward multimodal analysis. A survey by Liyanarachchi et al. (2025) methodically reviews the advanced state in Multimodal Music Emotion Recognition (MMER). The paper outlines a standard four-stage framework for MMER systems: (1) data selection, (2) feature extraction, (3) feature processing, and (4) emotion prediction. It details the field's progression from traditional machine learning to modern deep learning architectures, emphasizing the importance of feature fusion for combining modalities like audio and lyrics. The survey identifies critical research gaps, most notably the lack of large-scale, culturally diverse annotated datasets. The presented framework provides a standard methodological pipeline for research in this area. Moreover, the identified need for more diverse datasets validates the contribution of a study focused on a specific musical culture, positioning the analysis of Filipino music as a direct response to a recognized gap in the field.

As such, central to the MMER framework is the critical stage of feature fusion, where data from different modalities are combined. Investigating the most effective ways to perform this integration, a study by Xie (2025) provides a direct comparison of various fusion methods for combining speech and text data in emotion recognition. Using the IEMOCAP dataset, the research evaluates and contrasts the performance of early, late, and hybrid fusion models. The core problem addressed is identifying the most effective strategy for integrating features from these two distinct modalities to achieve higher accuracy. The findings reveal that bimodal audio-text models significantly outperform single-modality models. More specifically, the study highlights that a hybrid fusion model using a deep neural network achieved the highest accuracy, indicating that a refined integration strategy is essential for capturing the complex connection between vocal tones and lyrical content. This is relevant to the present study as it provides a strong rationale for the bimodal approach and offers a comparative baseline for different fusion techniques that may be considered when building the model. It addresses the problem of how to best combine the audio and lyrics data for optimal performance.

Building on the importance of effective fusion strategies, researchers like Faria et al. (2024) examine the integration of speech emotion recognition (SER) and text-based sentiment analysis to develop a more comprehensive model of affective communication. The study addresses the challenge that vocal expressions (emotion) and the literal meaning of words (sentiment) can sometimes be misaligned. The authors propose a multimodal framework that uses a wav2vec2 model for audio feature extraction and a BERT-based model for text analysis, employing a cross-attention mechanism to fuse the features. Their results show that this fusion approach leads to a richer and accurate classification of affective states than using either modality alone. This research is relevant to the project as it analyzes the precise audio-textual fusion task being undertaken. The use of modern transformer-based architectures (wav2vec2 and BERT) delivers a blueprint for a state-of-the-art methodology. It addresses the problem of conforming potentially conflicting signals between sung melodies and their lyrics, which is an occurrence in emotionally complex music.

Moreover, the scope of multimodal analysis can extend beyond audio and text and can incorporate other modalities. With this, Lian et al.'s (2023) survey presents a comprehensive review of Multimodal Emotion Recognition (MER) that utilizes deep learning-based techniques. The paper illustrates how MER plays a vital role in human-computer interactions (HCI) by identifying human emotions through combining signals extracted from speech, text, and facial-cue modalities. Their paper addresses the lack of studies on deep-learning-based MERs and critiques the various methods used in recent innovations for extracting emotional features. For instance, their paper provides a detailed analysis of different MER algorithms and highlights different fusion strategies such as early, late, hybrid, and intermediate layer fusion. Their evaluation and review of MERs and the different strategies employed provide a guide for future research to adequately select appropriate techniques for future MER development and HCI research.

In response to the challenges outlined in such surveys, researchers have proposed new approaches to advance the field. Kulkarni et al.'s (2025) paper introduces a novel deep-learning model for MER. Their model, the Empirical-Based Fusion Deep Convolutional Neural Network (EBF-DCNN), is designed to overcome the limitations of single-modality emotion recognition by combining audio, visual, and text-based features from video data. The model utilizes the Multimodal EmotionLines Dataset (MELD) and employs three distinct Deep Convolutional Neural Networks (DCNNs) to process each respective modality and fuses the outputs using an empirical-based fusion method, which minimizes overfitting and improves recognition speed. Their model was able to demonstrate better performance compared to other existing methods, achieving high rates of accuracy, precision, and recall.

The common step among all these advanced models is the fundamental process of feature extraction and data preprocessing, which is crucial for model performance. Before fusion can occur, raw signals must be effectively processed. Humans can easily determine which sounds are most useful, such as speech and music. In contrast, machines struggle to perform tasks that humans do effortlessly. The paper by Krishnan et al. (2020) delves into different techniques to help machines classify sounds into speech, music, and environmental noise as part of feature extraction. Each sound can be differentiated by its respective frequency range. However, environmental sounds are spread over the whole audible range, which makes it hard to determine. Data preprocessing, such as noise reduction and normalization, helps with this problem as it lessens or eliminates environmental noise.

The importance of these preprocessing steps, such as noise reduction, is demonstrated in practical applications where efficiency and accuracy are paramount. Built on the previously discussed concepts, a study by Hao et al. (2025), introduces a real-time multimodal music emotion recognition model that combines audio, visual, and physiological data using bidirectional long short-term memory (Bi-LSTM) networks and adaptive feature fusion. They found that dynamic sampling, noise reduction, and synchronized data alignment effectively boost both accuracy and system speed. Their pipeline demonstrates that targeted data cleaning and adaptive handling lead to robust and efficient emotion recognition, with results on benchmark datasets reflecting improved accuracy and lower latency compared to existing methods.

III. Methodology

To establish a quantitative baseline for emotional tonality, the original commercial recordings of the selected songs first underwent an initial acoustic analysis using the Librosa Python library. This assessment extracted key descriptors—specifically Tempo (BPM), Spectral Centroid, RMS Energy, and Zero-Crossing Rate—to define the ground truth values for Valence and Arousal that the model aimed to approximate. Subsequently, a dataset of twenty audio samples of Filipino singing was collected to serve as the experimental input. These recordings were captured using the Voice Memos application on a MacBook Air M1 within the hallways of the 6th floor of the Enrique Razon Sports Center in Malate, Manila, creating a dataset reflective of a natural, real-world acoustic environment. From this collection, two distinct datasets were constructed: one comprising the raw recordings and another consisting of audio processed for quality enhancement.

The preprocessing workflow for the enhanced dataset utilized FL Studio 25 in conjunction with the Edison plugin to perform targeted noise reduction and spectral cleaning. To further optimize the audio for emotion analysis, the FabFilter suite was employed, specifically using the Pro-Q equalizer with the "Vocal Lead 2" preset to excise harsh frequencies, and the Pro-C compressor with the "Upfront" preset to normalize dynamics. This ensured that the processed dataset achieved a consistent level of clarity and dynamic range suitable for accurate emotion recognition. To address the limited dataset size of twenty audio recordings, each audio file was processed with a maximum duration of thirty seconds using librosa's audio loading functionality. This approach optimized processing efficiency while maintaining the integrity of the original recordings.

Both raw and preprocessed audio files were analyzed to extract audio features. The audio feature extraction process captured various characteristics including Mel-frequency cepstral coefficients (MFCCs), chroma features, mel spectrograms, spectral properties (centroid, rolloff, bandwidth, contrast), tempo, energy measurements (RMS statistics, energy dynamics), pitch characteristics (fundamental frequency statistics), harmonic and percussive components, and Tonnetz features. For the lyrics, text features were extracted using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization with bigram support, capturing word patterns and relationships. Additional linguistic features were included such as exclamation and question mark ratios, word count, average word length, and capital letter usage, which can indicate emotional intensity in the text.

The emotion recognition system used traditional machine learning models suitable for small datasets: Random Forest and Support Vector Machine (SVM) classifiers. These models were chosen because they perform well with limited training data. The audio and text features were combined by concatenating them into a single feature vector, allowing the models to learn from both the acoustic properties of the music and the semantic content of the lyrics simultaneously.

Model evaluation was conducted using Leave-One-Out Cross-Validation, which is ideal for small datasets. This method trains the model on 19 samples and tests on the remaining sample, repeating this process 20 times so that each sample serves as the test case once. This approach maximizes the use of available data while providing reliable performance estimates. To thoroughly assess whether audio preprocessing improved the model's performance, three different feature configurations were evaluated: audio features alone, text features alone, and combined audio and text features. This allowed for a comprehensive comparison of how preprocessing affects different aspects of emotion recognition.

Statistical analysis was performed to determine if the differences between raw and preprocessed audio performance were significant. A paired t-test was used to compare accuracy scores, and McNemar's test was used to evaluate classification agreement between the two approaches. These tests provided statistical evidence for whether the audio enhancement workflow significantly improved the model's ability to predict emotions. The model's predictions were also compared against two baselines: the initial acoustic analysis results and the authors' interpretation of emotion, providing insight into how well the model aligns with different reference points.

IV. Results, Analysis, and Discussion

A comparative review reveals notable differences between the initial acoustic baseline established by Librosa and the authors' own labeling of the songs. While the automated analysis relied exclusively on technical aural features such as tempo and spectral brightness, the authors' interpretation incorporated the actual meaning of the lyrics, leading to conflicting classifications. A prime example of this is Ben&Ben's Kathang Isip; although the Librosa model classified the track as "Happy" due to its upbeat rhythm and major key tonality, the song is generally regarded as a sad anthem about heartbreak (Billboard Philippines, 2018).

This instance highlights a prevalent paradox within the Original Pilipino Music (OPM) genre, where artists frequently employ the technique of singing sad lyrics over an otherwise happy beat. This inherent contrast between the audio features and the lyrical sentiment makes it especially hard to identify the true emotion using just audio analysis. This validates the study's decision to use a multimodal approach, proving that integrating textual analysis is necessary to resolve the ambiguity between a song's sound and its true emotional intent.

Based on the data presented in Table 2, a distinct divergence is observable between the classifications derived from the initial acoustic analysis and the authors' interpretation compared to the outputs of the Random Forest models. This discrepancy can be largely attributed to the source material used for each assessment. The initial acoustic analysis was predicated on the original commercial recordings, which feature full instrumental arrangements. The presence of these instrumental elements—such as driving percussion or bright synthesizers—significantly influences the extracted spectral and energy features, often inflating valence and arousal scores.

In contrast, the Random Forest models were trained on natural singing samples (limited to 30-second durations) which lacked this complex instrumental backing. Consequently, songs originally classified as "Happy" due to their high-energy instrumentation (e.g., Ang Huling El Bimbo) were reclassified as "Sad" by the models. Without the accompanying instrumentation to drive the acoustic intensity, the models relied primarily on the isolated vocal features—such as pitch characteristics and harmonic components—where the underlying melodic melancholy became the dominant trait.

Notably, the results indicate that the Authors' Interpretation aligned most closely with the Combined Random Forest Analysis, particularly when applied to the Preprocessed Audio dataset. For instance, in tracks such as Ang Panginoon Ang Aking Pastol, Gitara, and Nais Ko, the Raw Audio model misclassified the emotion (often as Happy or Sad), whereas the Preprocessed model correctly aligned with the authors' "Neutral" classification. Similarly, for Bukas Na Lang Kita Mamahalin, the Raw model predicted "Anger," while the Preprocessed model converged with the authors' "Sad" interpretation. This stronger alignment suggests that the targeted noise reduction and dynamic normalization (via FabFilter Pro-Q and Pro-C) successfully removed artifacts that were confusing the classifier. Furthermore, the concatenation of TF-IDF text features allowed the model to leverage semantic context, mirroring the human authors' perception which inherently considers both the clarity of the vocal delivery and the lyrical meaning.

Figure 3 illustrates the comparative performance of Random Forest and Support Vector Machine (SVM) classifiers across four key metrics: Accuracy, Precision, Recall, and F1-Score. The charts highlight that the Random Forest model (green bars) achieved higher performance across all metrics when using the preprocessed dataset compared to the raw dataset (blue bars), whereas the SVM model maintained consistent performance regardless of the preprocessing.

Metric	**Random Forest**	**Support Vector Machine (SVM)**

Accuracy (Raw Audio)	0.3500	0.4000
Accuracy (Preprocessed)	0.5000	0.4000
Difference	+0.1500	0.0000
Accuracy p-value *(Paired t-test)*	0.1864	N/A
Significance (Accuracy)	No	No
McNemar p-value	0.3750	1.0000
Significance (McNemar)	No	No

Table 3. Statistical Comparison of Model Performance between Raw and Preprocessed Audio Datasets

Table 3 presents the statistical evaluation of the Random Forest and Support Vector Machine (SVM) classifiers when trained on raw versus preprocessed audio features. The results indicate a notable improvement in the Random Forest model, which exhibited a 15% increase in accuracy (from 35% to 50%) after the dataset underwent noise reduction and dynamic normalization via the FabFilter suite. In contrast, the SVM model showed no variance in performance, maintaining a static 40% accuracy across both experimental conditions.

However, despite the observable performance lift in the Random Forest classifier, the statistical tests reveal that these improvements are not statistically significant. The paired t-test yielded a p-value of 0.1864 for accuracy, and McNemar's test produced a p-value of 0.3750, both of which exceed the standard significance threshold (α = 0.05). Consequently, the null hypothesis—that preprocessing has no effect on model performance—cannot be rejected based on this specific dataset.

The lack of statistical significance can be primarily attributed to the constraints of the dataset size (n = 20). Even with the robust Leave-One-Out Cross-Validation method, the small sample size results in wide confidence intervals, meaning that even a substantial 15% improvement in accuracy lacks the statistical power to be deemed non-random.

Furthermore, the models' difficulty in achieving higher accuracy may stem from the inherent complexity of the OPM genre compared to the standard datasets typically used in Music Emotion Recognition (MER) research. Most emotion recognition algorithms and feature sets (such as standard TF-IDF vocabularies or western-centric tonality features) are optimized for Western music and English linguistic patterns (Soleymani et al., 2013). These traditional models often struggle to decode the specific cultural nuances of OPM—specifically the "sad lyrics, happy beat" paradox. The feature vectors extracted (e.g., TF-IDF bigrams and MFCCs) might lack the contextual depth required to resolve the dissonance between the upbeat acoustic features and the melancholic Filipino lyrics without a significantly larger training set to learn these complex non-linear relationships.

Figure 4 details the specific classification behaviors of the Random Forest model, providing granular insight into why the accuracy improved from 35% to 50% after preprocessing.

In the Raw Audio matrix (left), the model exhibits a severe class imbalance bias, overwhelmingly predicting "Sad" for the majority of inputs regardless of their actual label. Specifically, while it correctly identified 7 "Sad" songs, it completely failed to classify "Anger" (0 correct) and "Happy" (0 correct), and misclassified the vast majority of "Neutral" and "Anger" tracks as "Sad." This suggests that in the raw recordings, the ambient noise or lack of dynamic clarity caused the model to default to the majority class or the class with the most prominent low-frequency features (Sadness).

The Preprocessed Audio matrix (right) reveals where the audio enhancement made the most significant impact. The most notable improvement is in the "Neutral" category; whereas the raw model failed to identify any "Neutral" songs correctly, the preprocessed model correctly classified 3 out of 5. This indicates that the spectral cleaning and dynamic normalization helped the model distinguish the subtle, low-arousal characteristics of "Neutral" songs from "Sad" ones. However, the model continues to struggle with high-arousal emotions ("Happy" and "Anger"), frequently misclassifying them as "Sad" or "Neutral." This persistence of error for high-energy tracks reinforces the earlier discussion regarding the OPM paradox—where upbeat acoustic features in "Happy" or "Angry" songs may be semantically undercut by sad lyrics, confusing the multimodal classifier.

V. Conclusion and Final Work

This study investigated the impact of audio preprocessing on bimodal Music Emotion Recognition (MER) systems within the specific cultural context of Original Pilipino Music (OPM). By analyzing twenty natural singing recordings processed via FL Studio 25 and the FabFilter suite, we were able to unveil the unique "OPM paradox," a stylistic tendency to pair upbeat instrumentation with melancholic lyrics, which often confounds unimodal acoustic classifiers.

The results provided empirical evidence that preprocessing is a critical step for systems trained on limited, real-world data. The Random Forest classifier demonstrated a tangible performance boost, increasing accuracy from 35% to 50% after noise reduction and spectral balancing. This improvement was particularly evident in the retrieval of "Neutral" tracks, where artifact removal allowed the model to better parse subtle vocal cues that were previously obscured in the raw audio. In contrast, the Support Vector Machine (SVM) proved unresponsive to these enhancements, suggesting that tree-based ensembles may be more suitable for small, high-dimensional singing datasets.

However, while the gains in accuracy were observable, statistical validation through paired t-tests and McNemar's tests revealed that these improvements were not statistically significant ($p > 0.05$). This result is primarily driven by the constraints of the dataset size ($n=20$) and the high variability of natural singing. Consequently, while we established that the feature concatenation strategy (late fusion) of TF-IDF and Librosa features is a viable baseline, the null hypothesis regarding the statistical significance of preprocessing cannot be rejected without a larger corpus. Ultimately, this study highlights that while clean audio is necessary for "human-aligned" ground truth, it must be supported by more complex fusion strategies to fully capture the emotional intricacies of Filipino singing.

Thus, with these findings in consideration to further the development of culturally adaptive and accurate MER systems, future research could explore and prioritize the following directions. First, to overcome the statistical limitations identified in this study, future work must significantly expand the dataset size beyond 20 samples. Increasing the volume and diversity of OPM sub-genres is essential to narrow confidence intervals and improve the model's generalization capabilities. Additionally, while this study utilized a late fusion approach, future models should explore hybrid or cross-attention fusion mechanisms. These architectures can better model the non-linear relationships between modalities, effectively "weighing" the sadness of lyrics against the happiness of the beat to resolve the OPM paradox.

Moving beyond Random Forest and SVM, future research could also adopt Deep Neural Networks, specifically using Convolutional Neural Networks (CNNs) for Mel-spectrogram analysis and Transformer-based models for more nuanced lyrical sentiment extraction. Likewise, future iterations should incorporate additional modalities such as facial expressions or physiological signals for better empathetic systems. These signals could serve as a more objective ground truth to support self-reported emotion labels. Building on the preprocessing pipeline established here, future development should aim for real-time empathic music applications. This includes adaptive karaoke systems or therapeutic tools capable of processing live singing input and responding to the user's emotional state instantaneously.

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VII. AI Use Declaration

The Google Gemini Pro 2.5 Model was used by the researchers to improve the writing style and grammar of this paper. The initial contents, however, were entirely created by the researchers and edits made by the AI were manually checked and validated. The same model was also used to assist in the development of the model and statistical analysis techniques in Google collab.

VIII. Appendix

Song Title	Artist	Initial Acoustic Analysis Result	Authors' Interpretation
Ang Huling El Bimbo	Eraserheads	Happy	Sad
Ang Panginoon Ang Aking Pastol - Salmo 23	Jesuit Music Ministry	Neutral	Neutral
Bituing Walang Ningning	Sharon Cuneta	Sad	Sad
Bukas Na Lang Kita Mamahalin	Lani Misalucha	Sad	Sad
Di Na Muli	The Itchyworms	Sad	Sad
Gitara	Parokya Ni Edgar	Happy	Neutral
Huwag Muna Tayong Umuwi	BINI	Happy	Sad
Huwag Na Huwag Mong Sasabihin	Kitchie Nadal	Angry	Angry
Ikaw	Sharon Cuneta	Happy	Happy
Ikaw	Yeng Constantino	Happy	Happy
Kathang Isip	Ben&Ben	Happy	Sad
Kisapmata	Rivermaya	Happy	Sad
Lipad Ng Pangarap	Dessa	Happy	Neutral
Migraine	Moonstar88	Happy	Sad
Minsan Lang Kita Iibigin	Ariel Rivera	Happy	Neutral
Nais Ko	Basil Valdez	Happy	Neutral
Ngayon At Kailanman	Basil Valdez	Happy	Happy
Sana Maulit Muli	Gary Valenciano	Sad	Sad
Sino Nga Ba Siya	Sarah Geronimo	Sad	Angry
Upuan	Gloc-9	Angry	Angry

Table 1. Comparison of Emotional Classifications between Initial Acoustic Analysis and Authors' Interpretation

Song Title	Initial Acoustic Analysis Result	Authors' Interpretation	Combined (audio and text) Random Forest Analysis of Raw Audio	Combined (audio and text) Random Forest Analysis of Preprocessed Audio
Ang Huling El Bimbo	Happy	Sad	Sad	Sad
Ang Panginoon Ang Aking Pastol - Salmo 23	Neutral	Neutral	Happy	Neutral
Bituing Walang Ningning	Sad	Sad	Sad	Anger
Bukas Na Lang Kita Mamahalin	Sad	Sad	Anger	Sad
Di Na Muli	Sad	Sad	Neutral	Neutral
Gitara	Happy	Neutral	Sad	Neutral
Huwag Muna Tayong Umuwi	Happy	Sad	Sad	Sad
Huwag Na Huwag Mong Sasabihin	Angry	Angry	Sad	Sad
Ikaw	Happy	Happy	Neutral	Neutral
Ikaw	Happy	Happy	Neutral	Neutral
Kathang Isip	Happy	Sad	Sad	Sad
Kisapmata	Happy	Sad	Sad	Sad
Lipad Ng Pangarap	Happy	Neutral	Sad	Sad
Migraine	Happy	Sad	Sad	Sad
Minsan Lang Kita Iibigin	Happy	Neutral	Sad	Sad
Nais Ko	Happy	Neutral	Sad	Neutral
Ngayon At Kailanman	Happy	Happy	Sad	Sad
Sana Maulit Muli	Sad	Sad	Sad	Sad
Sino Nga Ba Siya	Sad	Angry	Sad	Sad
Upuan	Angry	Angry	Sad	Sad

Table 2. Comparison of Emotion Classification Results across Initial Analysis, Authors' Interpretation, and Random Forest Models using Raw and Preprocessed Audio

```python
def analyze_audio_file(file_path):
    print(f"Analyzing {os.path.basename(file_path)}...")

    # 1. Load the audio file
    # y = audio time series, sr = sampling rate
    y, sr = librosa.load(file_path, duration=60) # Analyze first 60 seconds to save time

    # 2. Extract TEMPO (Correlates to Energy/Danceability)
    tempo, _ = librosa.beat.beat_track(y=y, sr=sr)

    # 3. Extract RMS ENERGY (Loudness/Intensity)
    rms = librosa.feature.rms(y=y)
    avg_energy = np.mean(rms)

    # 4. Extract SPECTRAL CENTROID (Brightness/Timbre)
    # Bright sounds (High Centroid) often correlate with Happiness or Anger
    # Dark sounds (Low Centroid) often correlate with Sadness
    cent = librosa.feature.spectral_centroid(y=y, sr=sr)
    avg_brightness = np.mean(cent)

    # 5. Extract ZERO CROSSING RATE (Noisiness)
    # High ZCR often means distortion/noise (Anger)
    zcr = librosa.feature.zero_crossing_rate(y)
    avg_zcr = np.mean(zcr)

    # --- NORMALIZE THE DATA (Convert to 0.0 - 1.0 scale) ---
    # These are rough normalization factors based on typical pop music
    norm_energy = min(avg_energy * 10, 1.0)
    norm_brightness = min(avg_brightness / 3000, 1.0)

    return tempo, norm_energy, norm_brightness, avg_zcr
```

Figure 1. Code Snippet of Program that Performs Initial Acoustic Analysis on Selected Songs

Figure 2. Audio Preprocessing Interface displaying FL Studio 25, Edison, and the FabFilter Suite

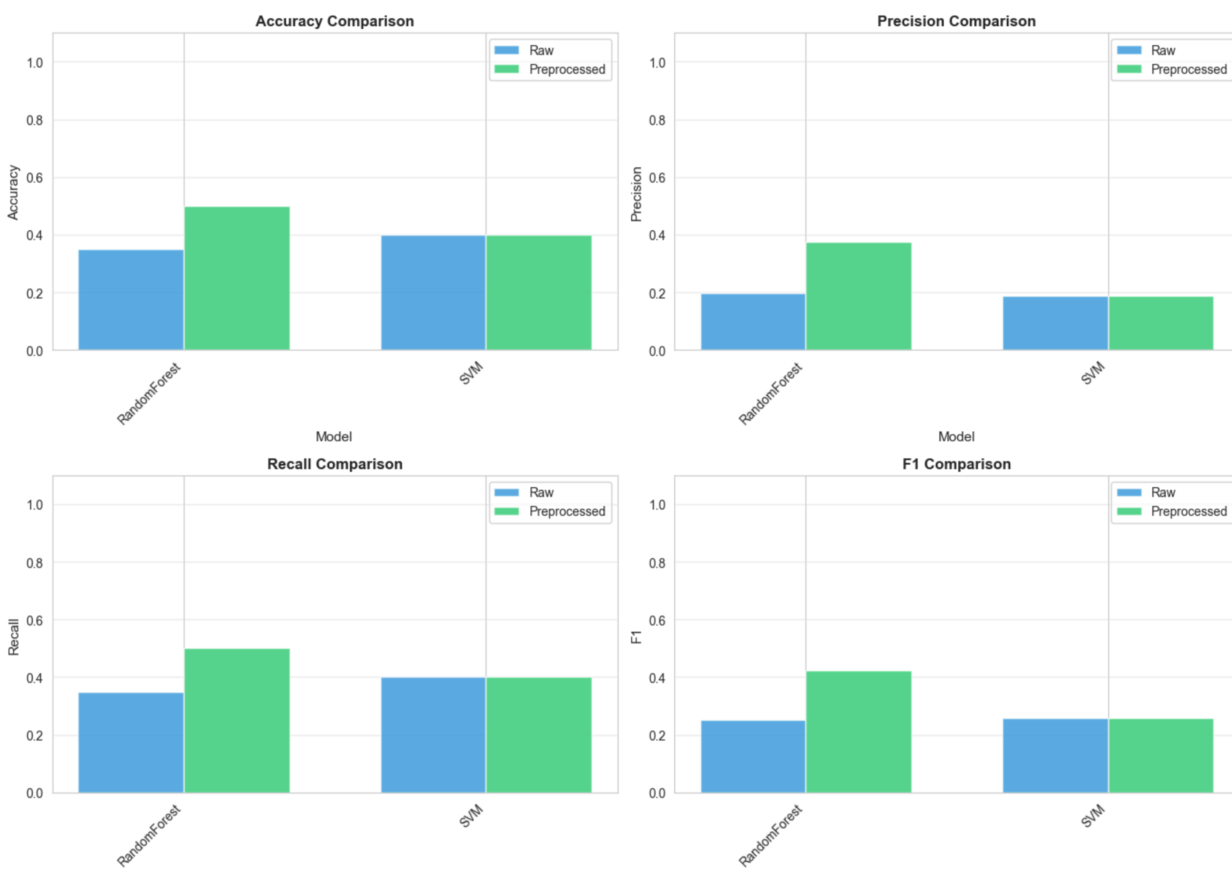

Figure 3. Performance Metrics Comparison of Random Forest and SVM Models on Raw versus Preprocessed Audio Data

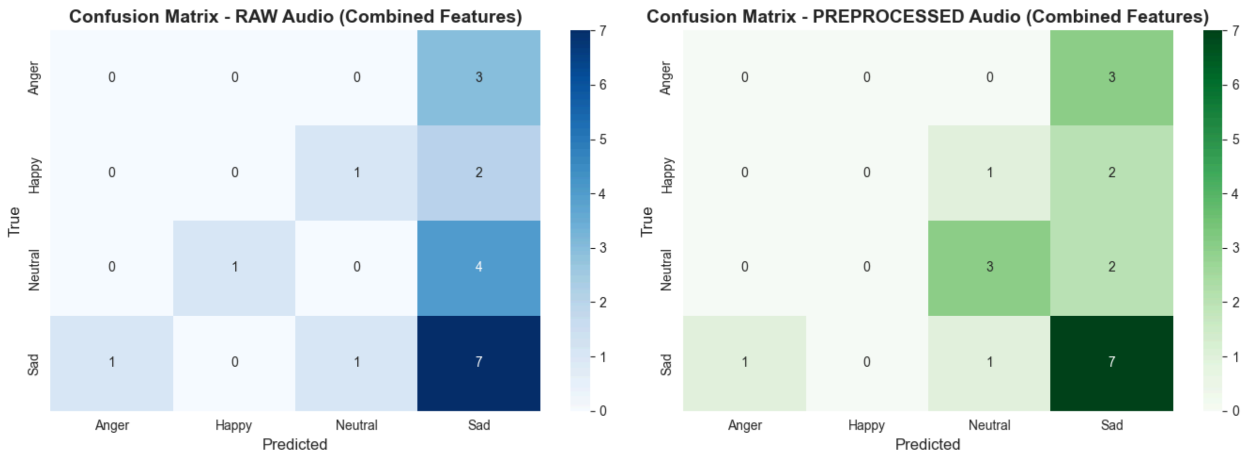

Figure 4. Confusion Matrices of the Random Forest Model on Raw versus Preprocessed Audio

Link to Notebook, Models, and Analysis:
[Emotion_recognition_notebook.ipynb]
Recording samples:
Samples